

PHYS 262 Capacitance Pre-lab

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The capacitance of a parallel plate capacitor is given by:

$$C^* = \epsilon_r \epsilon_0 \frac{A}{d} = \epsilon_r C,$$

where ϵ_0 is the permittivity of free space, ϵ_r is the relative permittivity of the dielectric material, A is the area of each plate, and d is the distance of separation. For paper, $\epsilon_r \approx 3.7$.

1. Calculate the capacitance for plates that are separated by $\frac{1}{4}$ inch of paper, and that have the area of an 8.5" $\times$ 11" piece of paper.

The area of the plates is $A = (8.5 \text{ in})(11 \text{ in}) = (0.2159 \text{ m})(0.2794 \text{ m}) = 0.06032 \text{ m}^2$.

The separation of the plates is $d = 0.25 \text{ in} = 0.00635 \text{ m}$.

The capacitance is:

$$\begin{aligned} C^* &= \epsilon_r \epsilon_0 \frac{A}{d} \\ &= (3.7)(8.85 \cdot 10^{-12} \text{ F m}^{-1}) \frac{0.06032 \text{ m}^2}{0.00635 \text{ m}} \\ &\approx 3.111 \cdot 10^{-10} \text{ F}. \end{aligned}$$

2. Calculate the capacitance if the area and separation are halved.

Since $C^* \propto \frac{A}{d}$, multiplying both A and d by the same factor $\frac{1}{2}$ will multiply C^* by a factor of $\frac{1/2}{1/2} = 1$. So, the capacitance will remain the same: $C^* \approx 3.111 \cdot 10^{-10} \text{ F}$.

Let there be two capacitors C_1 and C_2 , each with capacitance $1 \mu\text{F}$.

3. What is the equivalent capacitance of a series combination of C_1 and C_2 ?

The potential difference across C_1 is $\Delta V_1 = q/C_1$ and the potential difference across C_2 is $\Delta V_2 = q/C_2$.

So, the total potential difference across the series combination is:

$$\Delta V = \Delta V_1 + \Delta V_2 = \frac{q}{C_1} + \frac{q}{C_2} = q \left(\frac{1}{C_1} + \frac{1}{C_2} \right).$$

Assuming the equivalent capacitance of the series combination is C_{series} , then:

$$\begin{aligned} \Delta V &= \frac{q}{C_{\text{series}}} = q \left(\frac{1}{C_1} + \frac{1}{C_2} \right) \\ \frac{1}{C_{\text{series}}} &= \frac{1}{C_1} + \frac{1}{C_2}. \end{aligned}$$

4. What is the equivalent capacitance of a parallel combination of C_1 and C_2 ?

Since C_1 and C_2 are in parallel, they have the same potential difference: $\Delta V = \Delta V_1 = \Delta V_2$.

The total charge stored in the parallel combination is:

$$q = q_1 + q_2 = C_1 \Delta V + C_2 \Delta V = (C_1 + C_2) \Delta V.$$

Assuming the equivalent capacitance of the parallel combination is C_{parallel} , then:

$$\begin{aligned} q &= C_{\text{parallel}} \Delta V = (C_1 + C_2) \Delta V \\ C_{\text{parallel}} &= C_1 + C_2. \end{aligned}$$